Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *NLLB* | ***Context:*** *Amhara Region Agricultural and Microfinance Translation Cohort*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's output scoring ontology is a uniform scalar quality score (chrF++, spBLEU, or calibrated XSTS) applied to all directions and sentence types [Q455, Q648, Q661]. There is no domain-specific rubric for correctly rendering numerical eligibility conditions, conditional legal clauses, or financial terminology — the most consequential error types for the deployment. The XSTS scale measures 'meaning preservation' on a 5-point scale [Q654] but treats sentences uniformly. For toxicity, the taxonomy distinguishes source-present from added toxicity [Q705, Q706] with categories covering vulgar/profane/pornographic/hate-speech/bullying [Q712-Q714]; the paper acknowledges this is 'culturally sensitive' [Q715] with no Ethiopian regional calibration. A chrF++ improvement of +0.5 corresponds to detectable human improvement [Q624, Q631] but cannot distinguish silently altered subsidy thresholds from generally good prose. Independent 2025 evidence [WEB-10] shows mock translations reproducing named entities scored non-trivially on BLEU/ChrF++, confirming these metrics reward superficial overlap. Given OO is HIGH priority for binding documents, this is a structural-validity violation.

ID	Evidence
Q455	“We use the chrF++ metric to compare the model performance (see Section 7.1).” (p.53)
Q624	“We investigate the relationship between chrF++ score and human evaluation score to understand what quantity of automatic metric improvement in a model ablation overall, we find that chrF++ improvements of +0.5 chrF++ usually correlate to statistically significant human evaluation improvements, with a score of +1 chrF++ almost always being detectable by human evaluators.” (p.70)
Q631	“In conclusion, we believe that based on our human evaluation studies of model ablations, that an improvement of +0.5 chrF++ is often detectable by human evaluators.” (p.71)
Q648	“In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate.” (p.73)
Q654	“XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score.” (p.74)
Q661	“To obtain an aggregate human quality metric for each language direction in an evaluation study, we take the majority XSTS score for each sentence and average these majority scores over all evaluated sentences.” (p.74)
Q705	“In the context of translation, toxicity may originally be present in the source text or it can be generated de-novo in the target text (added toxicity).” (p.79)
Q706	“This added toxicity can come from a mistranslation (e.g., wrong lexical choice) or as a hallucination of a new target word from zero — both produce inaccurate translations.” (p.79)
Q712	“We include items that are commonly referred to as vulgar or profane language.” (p.80)
Q713	“In addition to these, we include items more specifically associated with depictions of pornographic content or sexual acts, some frequently used hate speech expressions, and some bullying expressions (including language that can cause trauma or be used with the purpose of silencing someone).” (p.80)
Q714	“We also include items, vulgar or not, referring to body parts that are commonly associated with sexual practices.” (p.80)
Q715	“Toxicity is culturally sensitive, which constitutes a challenge when starting from one source language (in this case, English) and attempting to find equivalents in such a largely multilingual setting.” (p.80)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Strengths

- XSTS protocol with calibration sets [Q650, Q664] addresses cross-language comparability of human evaluations, a non-trivial methodological contribution.
- Toxicity ontology distinguishing source-present vs. added toxicity [Q705, Q706] is conceptually appropriate for safety and partially relevant to the deployment's high-stakes context.
- The paper rejects unreliable model-based and roundtrip metrics [Q638, Q639] and explains its choice of chrF++ as the primary metric, providing transparency.

ID	Evidence
Q638	"While model-based metrics have shown better correlation with human judgment in recent metrics shared tasks (Freitag et al., 2021), they require training and are not easily extended to a large set of low-resource languages." (p.72)
Q639	"Another approach is to use highly approximate metrics based on roundtrip translations such as RttLangIDChrF (Bapna et al., 2022), although roundtrip translation may not correlate well with translation quality (Koehn, 2005)." (p.72)
Q650	"We use two advances — the XSTS evaluation protocol and the use of calibration sets — to enable meaningful human evaluation scores that are comparable across language pairs." (p.73)
Q664	"The calibration sets for all evaluation studies are drawn from a set of source, target sentence pairs consisting of K = 1000 backtranslated Flores-200 sentences of varying quality." (p.75)
Q705	"In the context of translation, toxicity may originally be present in the source text or it can be generated de-novo in the target text (added toxicity)." (p.79)
Q706	"This added toxicity can come from a mistranslation (e.g., wrong lexical choice) or as a hallucination of a new target word from zero — both produce inaccurate translations." (p.79)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The output label categories (chrF++, spBLEU, XSTS 1-5) [Q455, Q654, Q648] are general translation-quality scores. They are not designed to flag deployment-critical errors such as misrendered eligibility thresholds, altered penalty rates, or inconsistent terminology across clauses.

OO-2: Missing categories specific to the deployment: (a) numerical-fidelity scoring (correct rendering of percentages, dates, monetary amounts in legal clauses); (b) terminology-consistency scoring across a document; (c) conditional-logic preservation scoring; (d) administrative-register adherence scoring. None of these exist in NLLB's output ontology [Q455, Q648, Q654].

OO-3: The toxicity taxonomy is acknowledged as culturally sensitive [Q715, Q716] with English-anchored start [Q717], and no Ethiopian-specific calibration is documented. This may encode source-culture defaults but is a secondary concern relative to the missing domain-correctness rubric.

OO-4: Stakeholder-driven taxonomy redesign would be required: federal bureau translators (deployer's stated authority [elicitation A3]) would need to define a domain rubric distinguishing acceptable from catastrophic translation error types for subsidy/credit/land documents.

OO-5: Taxonomy issues harming validity: (a) structural validity — output ontology does not represent the construct of 'fitness for binding administrative documents'; (b) content validity — categories for high-consequence error types are absent; (c) external validity — chrF++ improvements do not predict deployment fitness [WEB-10].

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Q654	“XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score.” (p.74)
Q661	“To obtain an aggregate human quality metric for each language direction in an evaluation study, we take the majority XSTS score for each sentence and average these majority scores over all evaluated sentences.” (p.74)
Q715	“Toxicity is culturally sensitive, which constitutes a challenge when starting from one source language (in this case, English) and attempting to find equivalents in such a largely multilingual setting.” (p.80)
Q716	“Hate speech terms such as racial or ethnic slurs, for example, may well be the most challenging of all.” (p.80)
Q717	“We begin based on the professional translation of an initial list assembled in English, and allow additions to adapt to cultural specificities.” (p.80)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Information Gaps

- No documented case study of NLLB mistranslating a numerical legal-financial clause in Amharic specifically [WEB-10]; the gap is inferential from metric design rather than empirically demonstrated for this language.

Requires Expert Verification

- Federal bureau translators' specification of which error categories should be tracked for subsidy/credit document QA — required to design any deployment-specific rubric.

Remediation

Gap 1: Scoring is uniform scalar chrF++/spBLEU/XSTS [Q455, Q648, Q654] with no rubric for numerical-fidelity, conditional-logic preservation, or terminology consistency — the highest-consequence error types for binding documents.

Recommendation: Develop a deployment-specific error rubric (in collaboration with federal bureau translators) that flags catastrophic categories: (a) altered numerical thresholds or dates, (b) reversed/dropped conditional clauses, (c) inconsistent rendering of defined terms across the document, (d) loss of named entities. Score translations against this rubric in addition to chrF++.

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